From Cheap to Chic:
Enhancing Music Playback Quality of Budget Earphones via Hardware-Aware Learning

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INTRODUCTION

Low-end earphones are ubiquitous but deliver poor music quality due to hardware limitations.

46% of earphones cost less the USD 50

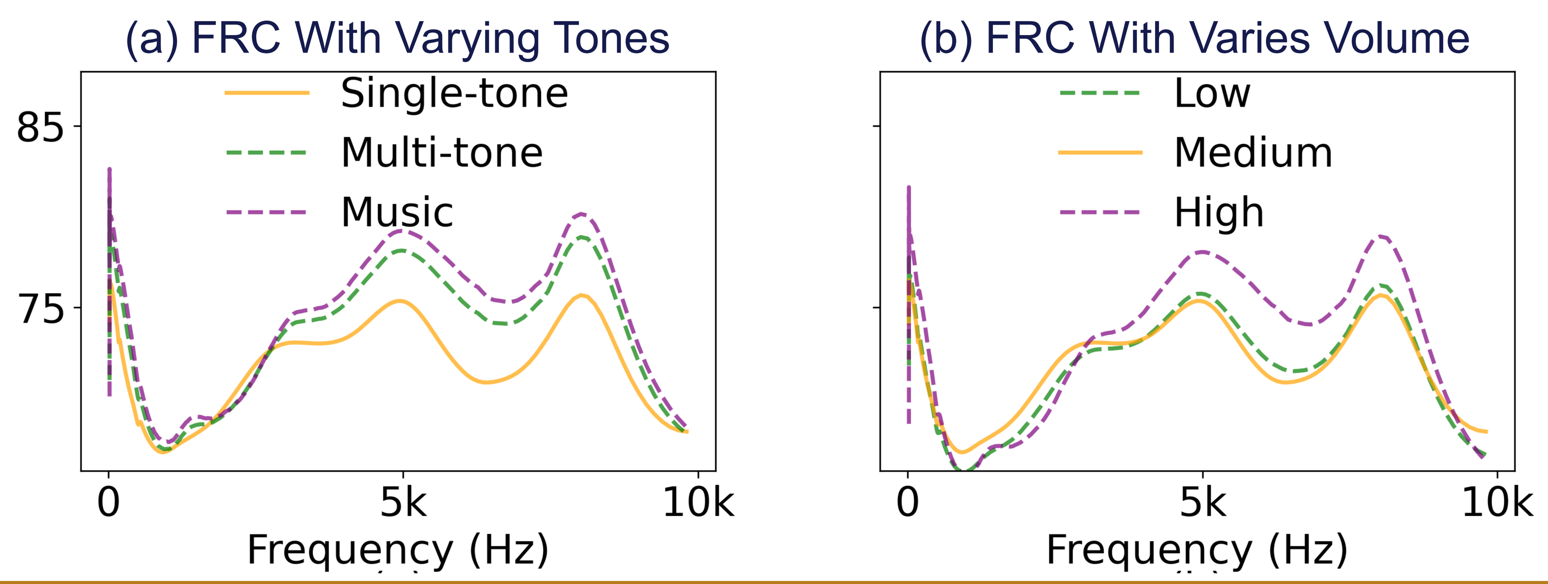
566M earphones shipped globally in 2024

EQ-based methods rely on frequency response curves (FRCs) measured under ideal conditions and fail to capture real-world non-linear distortion.

Our Goal: Learn to compensate hardware-induced distortions in the digital domain, without any hardware changes.

1. THE CHALLENGE

FRC is static and linear, while real hardware distortion is dynamic and nonlinear.



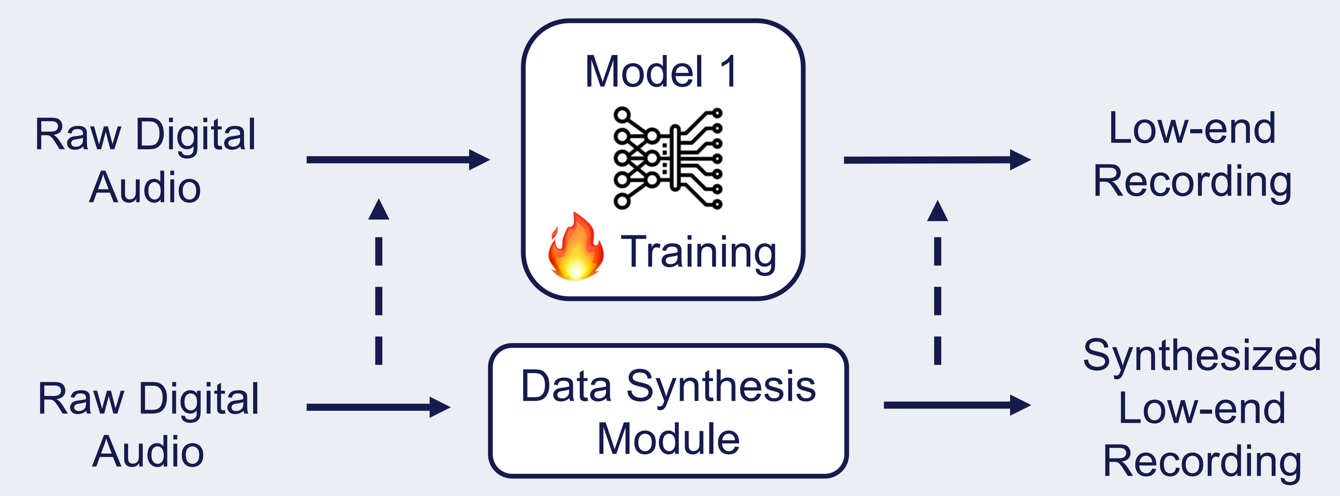
EQ Cannot Model:

- Harmonic Distortion
- Intermodulation Distortion
- Volume-Dependent Behaviors
- Complex Nonlinear Effects

2. CHEAP2CHIC: OUR APPROACH

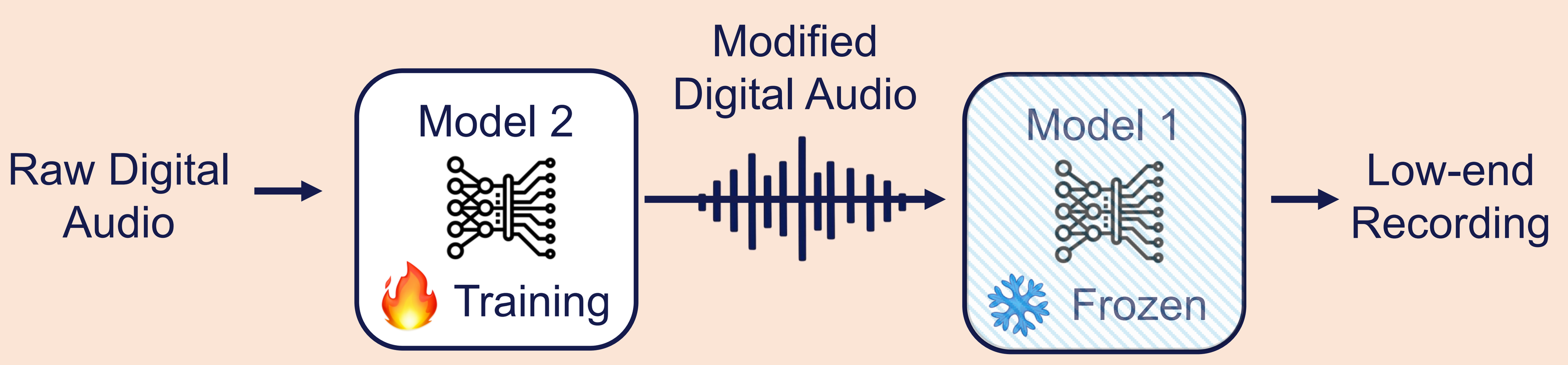
A two-stage framework that learns speaker characteristics and enhances audio to compensate hardware limitations.

Stage 1: Low-end Speaker Characterization



Learn a neural proxy that simulates how a low-end earphone distorts audio.

Stage 2: Digital Audio Enhancement

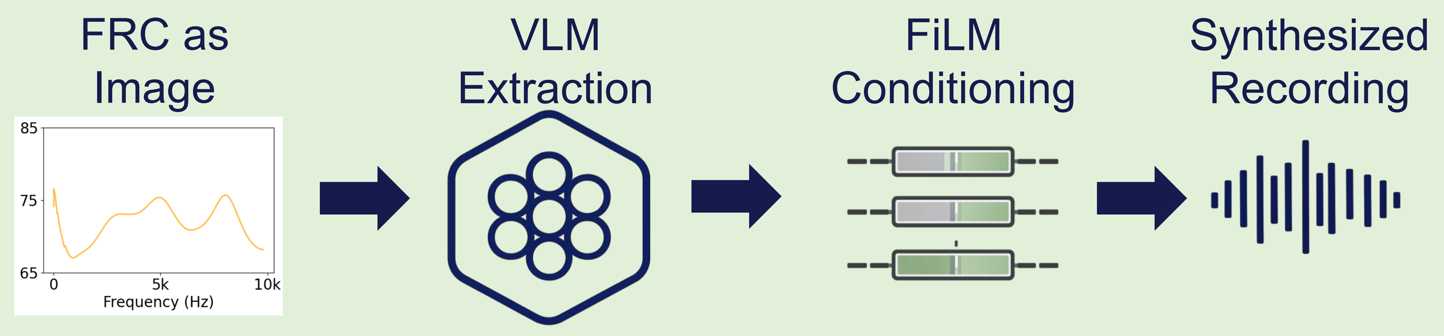


Learn to "pre-distort" the audio so that playback through the low-end device becomes high-quality.

Deployment: only Model 2 is required.

Data Synthesis Module

Reduce data collection from 180 min to 30 min



- Treat the long sequence FRC as an image
- Vision-Language Model extracts perception-aware features
- FiLM conditions the generative model

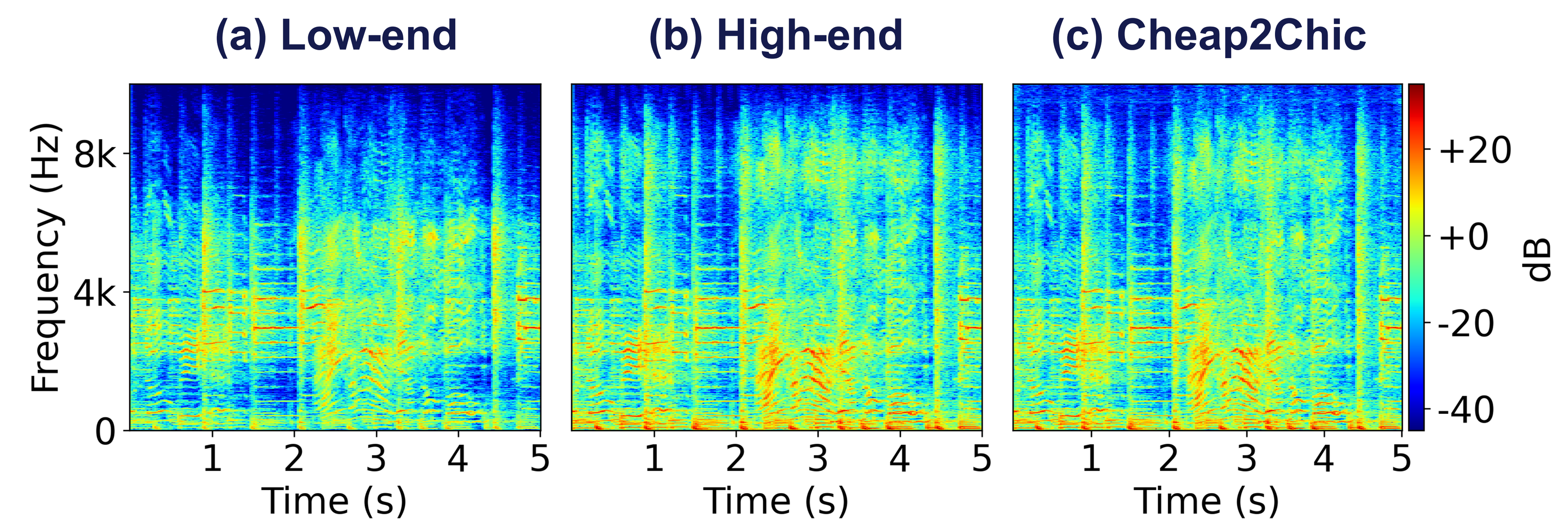
3. EVALUATION & RESULTS

3.1 Objective Quality (relative differences with high-end)

System	SI-SDR↑	PEAQ↑	LSD↓	FAD↓
Low-end Playback	0.05	2.43	7.41	1.40
AutoEQ	9.75	3.03	6.60	0.78
DeFTAN-II	16.18	3.22	4.71	0.64
HS-TasNet	17.20	3.68	4.49	0.43
Limited-RealData	16.64	3.35	4.52	0.41
Full-RealData	20.74	4.04	2.66	0.11
Cheap2Chic (Ours)	**20.56**	**3.91**	**2.93**	**0.14**

Cheap2Chic Significantly outperforms EQ-based and deep baselines across all metrics (+20 dB SI-SDR over low-end).

3.2 Spectrogram Comparison



Cheap2Chic recovers spectral details and harmonic structure close to high-end.

3.3 Devices Generalization

Intra-type Pairs		Cross-type Pairs	
In-ear → In-ear	Over-ear → Over-ear	In-ear → Over-ear	Over-ear → In-ear
+22.10 dB	+18.98 dB	+27.29 dB	+19.82 dB

3.4 Subjective Listening Test (Mean Opinion Score)

1 (Bad) – 5 (Excellent)

Low-end	**Cheap2Chic**	High-end
2.81	3.72	4.60

3.5 What Listeners Said

"Stronger bass and fuller sound."

"Instruments are clearer and more natural."

"Less harshness, more pleasant to listen."

3.6 Latency and Power Consumption

Cheap2Chic runs on a Xiaomi 13 smartphone, processing each 5-second audio segment in 1.83 seconds with 6% extra battery consumption per hour.

Why Cheap2Chic?

- Learns nonlinear Hardware behavior
- Efficient & real-time on mobile
- No hardware modification
- Work across content & volume

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